RAG Learning

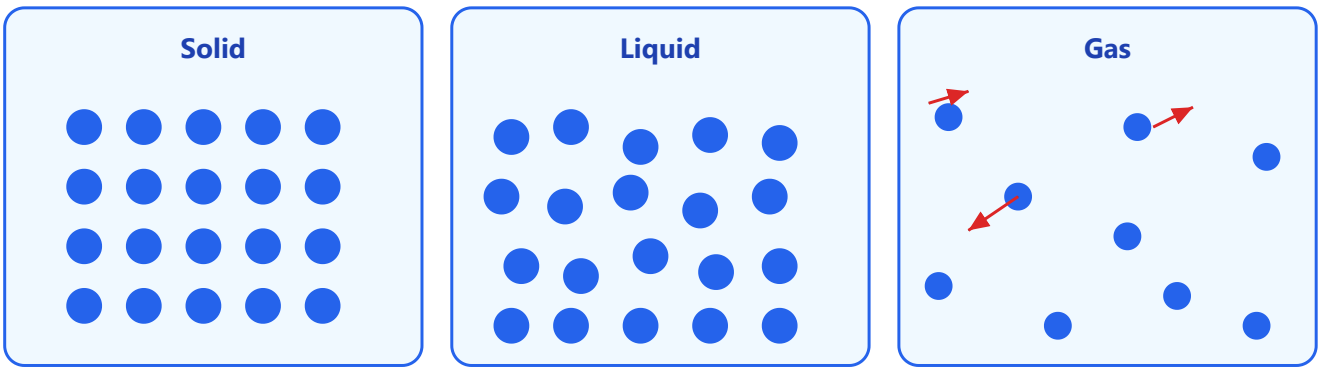
P6 — Matter: Models and Explanations

OCR GCSE Physics B (J259) — 21st Century Science · Paper 2 (Depth in Physics)

P6.1 The Particle Model of Matter

All matter is made of tiny particles (atoms/molecules) that are in constant motion. The **arrangement, spacing and motion** of these particles differ between solids, liquids and gases and explain the bulk properties of each state.

Particles in the Three States of Matter



regular pattern · touching · vibrating | random · touching · sliding | random · far apart · fast

P6.2 Density

Density (ρ) = mass per unit volume. Unit: **kg/m³** (or g/cm³).

$$\rho = m / V \quad (kg/m^3) = (kg) / (m^3)$$

Material	Typical density (kg/m^3)
Air (at 20 °C)	1.2
Cork	240
Ice	917
Water	1000
Sea water	1025
Aluminium	2700
Iron	7870
Gold	19 300

Worked Example: A metal block of mass 1.35 kg has volume 1.5×10^{-4} m^3. Find its density and suggest the material.
$\rho = m / V = 1.35 / 1.5 \times 10^{-4}$ = **9000 kg/m^3** → close to copper (8960).

Measuring Density

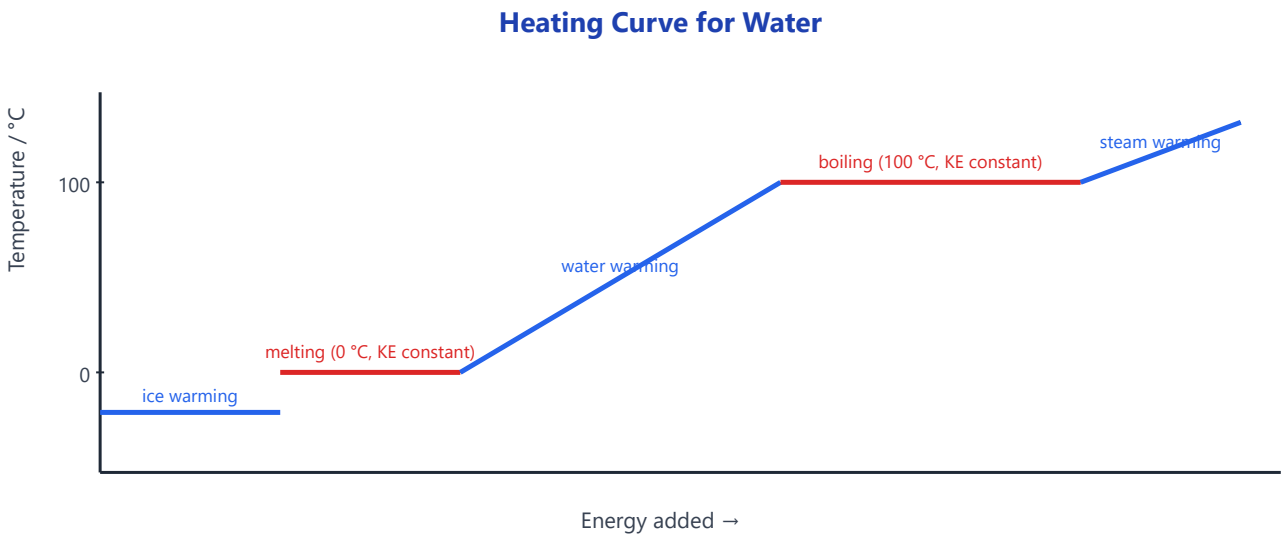
- **Regular solid**: measure mass with a balance; calculate volume from dimensions (l × w × h for a cuboid; $\pi r^2 h$ for a cylinder).
- **Irregular solid**: submerge in a **eureka/displacement can**; volume = volume of water displaced.
- **Liquid**: measure mass of empty measuring cylinder; add liquid; re-measure mass and volume directly.

P6.3 Changes of State and Internal Energy

The **internal energy** of a substance is the total energy of its particles — kinetic (from motion)

plus potential (from intermolecular forces).

- When a substance is **heated and not changing state**: kinetic energy rises → temperature rises.
- When a substance **changes state**: potential energy rises (bonds break) → temperature stays constant.
- Mass is **conserved** in a change of state — the same particles rearrange, no matter is gained or lost.



Specific Heat Capacity

Specific heat capacity (c) = energy needed to raise 1 kg of a substance by 1 °C. Unit: J/kg °C.

$$\Delta E = m \times c \times \Delta\theta$$

Substances with high c (like water, c = 4200 J/kg °C) take a lot of energy to warm up — which is why water is used in central heating systems and as a coolant.

Specific Latent Heat

Specific latent heat (L) = energy needed to change the state of 1 kg of a substance with no temperature change. Unit: J/kg.

$$E = m \times L$$

- **Fusion** (solid ↔ liquid): L_f for water ≈ 334 000 J/kg
- **Vaporisation** (liquid ↔ gas): L_v for water ≈ 2 260 000 J/kg

Worked Example: How much energy is needed to boil away 0.3 kg of water at 100 °C?
$E = m L_v = 0.3 \times 2\,260\,000 =$ **6.78×10^5 J**

P6.4 Pressure in Fluids

Pressure = force per unit area. Unit: **pascal (Pa)**. 1 Pa = 1 N/m².

$$p = F / A$$

In a fluid, pressure acts in all directions at a given depth. Pressure due to a column of fluid depends on depth and density:

$$p = h \times \rho \times g \quad (\text{Pa}) = (\text{m}) \times (\text{kg/m}^3) \times (\text{N/kg})$$

Pressure Increases with Depth

Upthrust (Archimedes)

When an object is in a fluid, the pressure on its lower surface is greater than on its upper surface. This gives a net upward force called **upthrust**. Upthrust = weight of fluid displaced.

- If upthrust > weight → object floats.
- If upthrust < weight → object sinks.
- Ships float because their shape displaces a large weight of water, even though their metal is denser than water.

P6.5 Atmospheric Pressure

The atmosphere is a fluid — a layer of air about 100 km thick. Air has weight. At sea level the pressure is about **1.0×10^5 Pa** (1 atmosphere, ~100 kPa). As altitude increases, there is less air above you, so atmospheric pressure falls.

This is why:

- Ears pop in aircraft — the external pressure changes and the pressure inside the middle ear equalises through the Eustachian tube.
- Boiling point of water falls at altitude — atmospheric pressure is lower, so it takes less molecular energy for water vapour to escape.

P6.6 The Gas Laws (qualitative)

At a fixed temperature, gas particles collide continuously with the container walls, exerting a pressure. This pressure changes if volume, temperature or amount of gas changes.

- **Temperature ↑ (constant V)**: particles move faster → more frequent/harder collisions → **pressure rises**.
- **Volume ↓ (constant T)**: same number of particles in smaller space → more frequent collisions → **pressure rises**. (Boyle's Law: pV = constant.)
- **Compressing a gas quickly** (e.g. a bike pump) does work on it → temperature rises → pump gets hot.

Boyle's Law: $p_1 V_1 = p_2 V_2$ (fixed temperature, fixed mass of gas)

Worked Example: A gas occupies 300 cm^3 at 100 kPa. What is its pressure when compressed to 120 cm^3 at constant T?
$p_2 = (p_1 V_1) / V_2 = (100 \times 300) / 120 =$ **250 kPa**

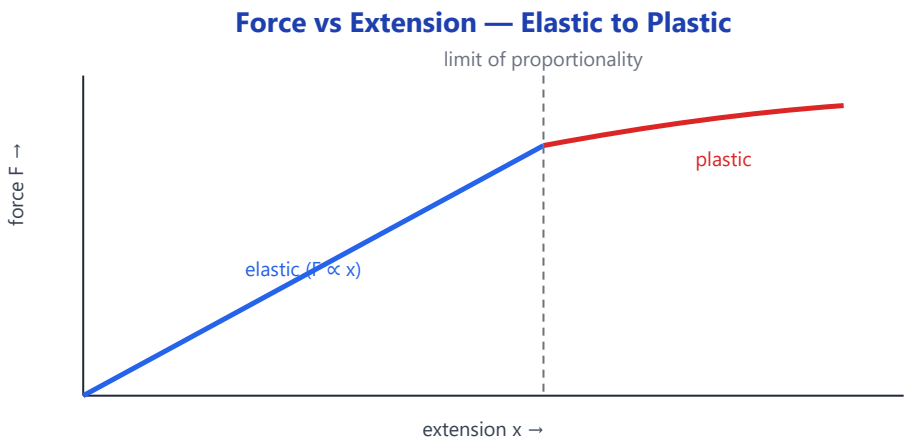
P6.7 Forces and Materials

Hooke's Law Recap (from P4)

For a spring in its elastic region: $F = k \times x$. The graph of force against extension is a straight line through the origin.

Elastic vs Plastic Deformation

- **Elastic deformation**: object returns to its original shape when the force is removed (below the limit of proportionality).
- **Plastic deformation**: object is permanently changed (above the elastic limit) — some of the work done is transferred to internal energy/heat.



P6.8 Summary & Key Equations

- $\rho = m / V$
- $\Delta E = m c \Delta\theta$ (heating/cooling)
- $E = m L$ (change of state)
- $p = F / A$
- $p = h \rho g$ (pressure in a fluid)
- $p_1 V_1 = p_2 V_2$ (Boyle's Law)
- $F = k x$ (Hooke's Law)

Exam Tip: Many students lose marks by confusing **weight** (a force, in N) and **mass** (an amount of matter, in kg), or **pressure** (N/m^2) with **force** (N). Always state units.

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